UAV as a Service: Enabling On-Demand Access and On-The-Fly Re-tasking of Multi-Tenant UAVs Using Cloud Services

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***Abstract*—As commercial roles for Unmanned Aerial Vehicles (UAVs) become clearer and demand for services provided by them increases, UAVs rely more on new cloud computing services and cooperative coordination to provide mission planning, control, tracking and data processing. This article presents the UAV as a Service (UAVaaS) framework, which ports features commonly found in traditional cloud services, such as Infrastructure, Platform, and Software as a Service, to the domain of UAVs. This work aims to conceptualize and design UAVaaS for commercial applications. Specifically, a cloud-provided orchestration framework that allows multi-tenant UAVs to easily serve multiple, heterogeneous clients at once and automatically re-task them to users with higher priority, mid-flight, if needed. The outcome of this research aims to provide an introductory overview of UAVaaS, explain core protocols and network topologies, and identify key system components and requirements.**

I. Introduction

Unmanned Aerial Vehicles (UAVs) are rapidly changing the way the world conducts business. Devices once used as reconnaissance and attack infrastructure for military missions, have quickly evolved into practical solutions to a wide range of commercial and industrial problems. Today, it is not uncommon for UAVs to fly commercial operations including structural inspections, wildlife protection, emergency management, land surveying, real estate, film production, border patrol, and agriculture. These UAVs can also send data to a home station or a cloud for processing as they perform the assigned mission [1]. Research and development institutions, trying to stay ahead in a new competitive market, have tackled major challenges in performance, handling, safety, management and communication domains, in the efforts of attracting potential customers. Currently, the U.S. Federal Aviation Administration (FAA) is under pressure to develop and implement new rules and regulations to allow for safe operations of UAVs within the National Airspace System (NAS), indicating that a major influx is expected in the near future.

Unfortunately, as the technology and capability of next generation commercial UAVs improve, so does their corresponding price tag; and while immediate advantages and benefits over traditional solutions are observable, high upfront capital cost pose as deal breakers for small companies. Other costs will also be incurred as a consequence of conducting commercial UAV operations including FAA licensing and certification, high insurance premiums to cover potential damage and liabilities, cost of highly trained pilots and expensive maintenance fees.

This article introduces a new approach to the way companies acquire and fly UAVs for commercial operations that would otherwise create a financial burden. By installing UAV hubs in areas of high commercial demand, populating them with network-aware UAVs and connecting them to a cloud service provider, clients (companies) can temporarily connect to UAVs and use them as a ”pay-as-you-go” utility service similar to the Infrastructure, Platform and Software as a Service business models seen today. UAV as a Service (UAVaaS) is an orchestration framework that helps to coordinate the scheduling, reservation, tasking, re-tasking and control of UAVs as needed by companies located within available service area coverage. Also integrated is a data consumption and billing management system that allows customers to track, monitor, and pay usage charges as flight operations are flown on a periodic basis. UAVaaS eliminates the long waiting times for a rental UAV to arrive in the mail, or for an outsourced UAV provider to arrive at a company’s location of interest. This wait time is avoided because UAVs are deployed autonomously and will fly directly to a designated starting point where control is then handed over to pre-authorized controllers. UAVaaS providers take care of all overhead such as acquisition, maintenance, licensing and insurance of UAVs allowing customers to focus on the mission at hand.

II. Background

Commercial UAV technology has progressed significantly in the past 10 years. According to a 2013 study by the Association of Unmanned Vehicle Systems International (AUVSI), a total of 103,776 new jobs will be created by 2025 within the commercial UAV industry, having a $75 billion in national economic impact [6]. To mainstream the registration process for companies, the U.S. Federal Aviation Administration has released a Small UAS Notice of Proposed Rule Making (NPRM), which once completed would be the primary method

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to receive authorization to conduct non-recreational operations. Currently under the much slower Section 333 Exemption process, more than 3,615 petitions have been granted for current UAV service companies.

Current cloud-based UAV solutions already exist. Skyward Incorporated provides a central repository for regulators, insurers and managers to access need-to-know information about flight operations. Skywards cloud-based solutions comes complete with airspace mapping and visualization, flight planning and logging, and record management systems for maintenance, personnel and inventory tracking. Drone Deploy Inc. provides cloud-based software that allows automated command and control of UAVs through the cloud, and enables real time uploading and analyzing of data. Uploaded data is conveniently processed for aerial mapping, terrain modeling and crop health visualization. Airware Inc. develops proprietary, cloud-ready flight controllers that provide out of the box connectivity and extensive customizability for drones in a variety of applications. Features of Airware products include, cloud connectivity, user authentication, flight simulation, real-time flight optimization, automated take-off and landing, flight planning, exporting and much more.

Remote UAV control has also undergone major improvements that eliminates dependency to proprietary hardware. Traditional UAVs were flown using remote controllers that operate on agreed Remote Control(R/C) frequencies such as 2.4GHz and 5.8 Ghz ranges. Many of these controllers were tightly coupled to the UAVs that use them and usually require both devicese to be shipped together when a change of ownership has occured. While this approach is simplistic, the range of communication is limited by the signal strength and power of the sending and receiving antennas which restricts operations within line of sight (LOS). It is also easy for attackers to hijack or disable UAVs just by transmitting and receiving on the same frequency. Wi-Fi adhoc networks mitigated this problem by creating private, WPA2 protected wireless networks for UAVs and their respective controllers to communicate. This allowed for a wide range of Android and iOS remote control applications to be installed on smart devices which eliminated the need for proprietary hardware altogether. Research is now focused on the use of 4G LTE cellular networking technology to increase the communication range to commercial UAVs, however there are many security and regulator issues that still need to be solved. While satellite communication proves to be the most reliable method of long distance control, the technology is generally very expensive and is usually reserved for military operations.

III. CHARACTERISTICS

The UAV as a Service architecture enables customer access to multi-tenant UAVs. Multi-tenancy in this context refers to the capability of a single UAV to serve the demands of multiple customers, while still maintaining privacy of data and control. This means that UAVs can be re-tasked seamlessly without the need of explicit communication or handover between companies. Some features include ”pay as you go” billing (similar to Amazon Web Services), on demand scalability, dynamic flight assignment, policy-based access control, scheduling and reservation, on-the-fly re-tasking, and transparent maintenance and upgrade by trained ground crew personnel.

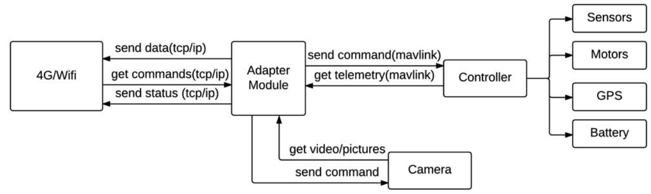

Fig. 1. Components of UAV subsystem

UAVaaS allows customers to perform any task through an internet browser, desktop, or mobile application that interfaces with cloud service API endpoints. These tasks include planning missions, spectating live video feeds and uploading waypoint information to UAVs. This approach eliminates the need for specialized ground control station equipment or proprietary Line of Sight (LOS) communication infrastructure. The primary flight maneuvers such as takeoffs, landings, waypoint navigation, speed and altitude changes are handled by the UAV’s on-board flight controller as illustrated in Figure 1. In order for these commands to be received over 4G LTE, a dedicated companion board, for example an Arduino or Rasberry Pi acts as an adapter that parses network packets into a protocol that the on-board flight controller understands. Multiple adapters can be designed to interface with different controller protocols, further reducing hardware and firmware dependency. Fail-safes such as preventing operators from disobeying speed and altitude restrictions, or intentionally trying to destroy the vehicle, is provided by additional software installed on the companion board. This is needed since companies will be in control of their entire operation instead of delegating this task to 3rd party UAV service providers.

All communication and tasking of UAVs to users are handled by a cloud coordinator service. This service is aware of all UAV information including flight capabilities, payload features, current tasking assignment, closest UAV hubs and much more, allowing it to make strategic and optimal decisions based on current demand.

IV. USER CATEGORIES

This research identifies five user categories who interact with UAVaaS. These categories are general operators, emergency services operators, spectators, ground crew personnel, and third party vendors. A description of each user category’s role is illustrated in Figure 2.

General operators are the primary users who require access to UAVs to satisfy business requirements. Operators plan and control flight operations, while spectators only require access to collected data (e.g., imagery). Emergency Service Personnel represent privileged higher priority organizations such as fire departments, police departments, and other emergency first

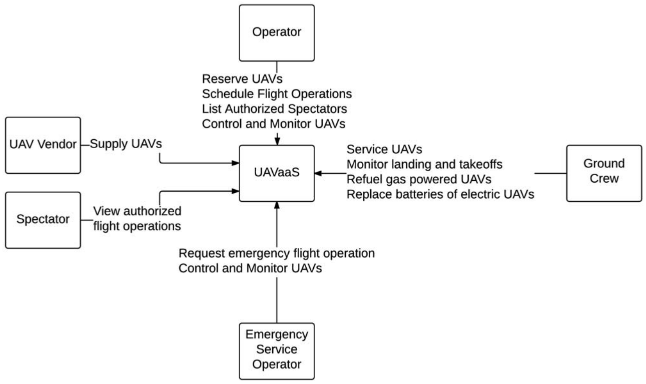

Fig. 2. User categories for the UAVaaS System

responders that request re-tasking of UAVs mid-flight or on ground for the purpose of using them in high-risk situations. The ground crew maintains and operates strategically located UAV hubs. Their role is to monitor the safety of all UAVs during take-off, while en-route, and landing as well as to respond to any accidents within specified service areas. UAV Vendors represent manufacturers who supply UAVs to the UAVaaS system.

The following describes an example that demonstrates the interaction of all user categories with the cloud coordinator service. Third party vendors manufacture and supply UAVs to a major UAVaaS hub due to its high population and commercial density. Ground crew personnel at these hubs, employed by the UAVaaS service provider, perform intake, registration and licensing operations and put UAVs online for immediate use. Fire fighters trying to extinguish a major forest fire contacts the cloud coordinator requesting a large number of UAVs to assist in reconnaissance, planning and detection of human life. Since the number of requested UAVs cannot be met as too many are already in use by agricultural, mining and construction zones, the cloud coordinator selects the most appropriate UAVs to retask (with current operator's permission). News agencies, disaster control, and government entities spectate live video feeds, overseeing the operation and assist in decision making.

V. Core Safety and Security Requirements

A set of core requirements and attributes are identified for overall safety and reliability of UAVaaS. These requirements target different aspects of the system that may be distributed across multiple components. These components must implement requirements in order to provide complete coverage and satisfy success criteria.

Data Integrity must be maintained with all aspects of communication and data storage operations in the UAVaaS network. This involves developing dynamic access control schemes and properly authorizing entities to access and modify data. Logging features should be incorporated to maintain non-repudiation and to handle accountability.

Data Confidentiality and Privacy must be maintained especially in a multi-tenant system, where the same storage devices and sensors are shared among multiple customers. Key management systems, encryption schemes and key exchange protocols ensure security of data and control commands. Distribution and revocation systems for encryption/decryption keys assist where UAVs are re-tasked to other companies or when spectators are only allowed to view data for a specified duration of time.

Session Hijacking Prevention entails preventing malicious users from taking control of an active session that can result in data being compromised and severe consequences to UAVs. These consequences include complete denial of service between customers and UAVs which would cause missions to automatically terminate, or sending malicious commands to UAVs that could disrupt customer operations such as turning off sensors and flying over unauthorized areas.

Efficient Scheduling assist in streamlining, as customers will most likely try to request access to UAVs during the same time slot. Therefore, there must exist an efficient scheduling algorithm to allow users to select and reserve UAVs and to notify appropriate spectators when a scheduled mission is canceled. Policies must be developed to ensure no denial of service can occur, such as in the case when a malicious user reserves UAVs for a period of time without the intention of flying the mission.

Efficient Re-tasking is required in emergency situations such as a natural disaster or terrorist attack. Emergency management services (EMS) may request UAVs that are in use by other companies to assist in their operations. Re-tasking algorithms should determine the most appropriate UAVs to retask based on mission parameters supplied by the EMS request. These parameters include details such as location of incident, estimated time of mission and payloads needed.

Smart Self-preservation mechanisms allow UAVs to detect and automatically prevent its own destruction caused by intentional or unintentional user commands and environmental situations. Users may for example try to fly beyond the capable range of the UAV's power supply or try to crash them into buildings or tall structures. UAVs may also experience weather conditions that may exceed aerodynamic flight envelopes. Smart systems should detect the occurrence of these actions and automatically terminate the mission and fly back to the nearest UAV hub.

Autonomous Contingency Management enables UAVs to become self aware of their current state and report any problems during or before flights. In the event of a power failure or mechanical malfunction, the UAVs have to be equipped such that they attempt to navigate and return to the nearest UAV hub. If a safe return is not possible, a distress message must be sent out to all UAV hubs with the last reported location, so that a ground crew recovery team can make an attempt to recover the device.

Situational Awareness mechanisms allow UAVs to detect FAA mandated no-fly zones such as airports and national parks and report to operators of impending unauthorized entry into those zones. Even when operators do not respond to warning messages sent by UAVs, an automatic override must be triggered to automatically avoid no-fly zones.

Scalability and Fault Tolerance is the UAVaaS capability of handling a large number of connections at scale while maintaining appropriate services levels. Too many UAVs can potentially be flown in a given service area leading to a bottleneck effect. The UAvaaS must provide load balancing or dynamic network routing through software-defined networks or other means to maintain quality of service and availability.

Send and Forget Memory Restrictions limit information storage on UAVs. This means that once data has been sent to UAVaaS cloud coordinators, it must be immediately erased or overwritten from memory. This prevents possible data leaks and potential forensics analysis to be performed to recover any confidential information. This requirements also reduces power consumption and weight of the UAV.

VI. System Design

UAVaaS is designed around a centralized cloud coordinator that acts as a marshaling service for all incoming and outgoing connections. This separates both users and UAVs by preventing direct access and line of communication to resources, and allows a non-biased entity to provide the multi-tenancy property. In real world implementations, multiple coordinator instances will be connected laterally to provide proper load balancing and to serve a range of geographical regions. Communication can be broken down into three message types: (1) Management, (2) Data, and (3) Control. The decision to decouple communication is made to improve ease of implementation for protocols and to simplify data prioritization and routing algorithms.

Figure 3 illustrates how controllers, spectators, ground crew personnel and UAVs use these messages and demonstrates the ease of scalability for additional UAVs and users. Management messages provide the coordination and administration properties of UAVaaS. Before any flight operation and any communication between customers and UAVs can begin, authorization and session negotiation must be handled first. Similarly, when flight operations are finished, termination of sessions, and disassociation of UAVs, controllers and spectators must be performed.

Control messages are relatively light-weight and provide controllers an interface to send control signals to the UAV. Since UAVs will be flying autonomously for the entire duration of the operation, messages will be simple control commands for altitude and speed changes, uploading new way points or points of interest, or controlling cameras and other payloads.

Data messages contain telemetry, video, pictures, warning logs or system messages sent back to the UAVaaS Coordinator which is then distributed to appropriate controllers and spectators where applicable. These messages may include high quality photos or videos that have been taken, or real-time altitude, speed and location information needed by the controller to make decisions about a current mission.

A. UAVaaS Coordinator Design

The UAVaaS Coordinator is designed with two key services that work together to satisfy all communication requirements. Two services are needed instead of just one because of the differing natures of management, control, and data messages. Management messages are sent and received as a request-response pair, meaning that when a request is received by the UAVaaS Coordinator, a response is sent back to the sender. Control and data messages work differently. Instead of the original sender receiving a response, the message is forwarded to its appropriate receiver(s). Management messages can therefore be processed using a simple REST service, and control and data messages can be processed using a forwarding/messaging service. Moreover, the messaging service is dynamically configured by the management service based on requests sent to it by UAVs or operators and spectators.

The management service handles the bulk of all coordination and scheduling. It sets up messaging sessions between operators, spectators and UAVs, and ensures only users scheduled for a given flight block may use them. It communicates with cloud storage databases to retrieve and create user account information and securely store collected data and mission logs for every flight flown. The management service also configures the messaging service based on the needs of operators and notifies ground crew of UAV issues such as warning, errors and battery status when grounded.

The messaging service handles duplication and forwarding of control and data messages between UAVs, operators and spectators, and error and warning log messages to the management service which ground crew personnel use to track and monitor UAVs in flight. A messaging service uses a publisher-subscriber architecture in which exchanges and queues work together to send and receive messages. Any subscriber interested in a messages type, for example logs, instantiates a queue and connect it to a data exchange. A publisher then simply publishes a message to the same data exchange flagged with the same topic specified by the subscriber.

Figure 4 illustrates communication between an Operator and two Spectators flying a mission with 3 UAVs. Operator A publishes a control to a control exchange Ctrl_Ex which duplicates and routes the messages to all intended UAV subscribers. UAVs in turn publish data to a data exchange Data_Ex which duplicates and routes the messages to all intended spectators and operators.

This architecture reduces additional overhead on UAVs that would need to expend computational resources to duplicate messages and send them directly to Operator A, Spectator A and Spectator B separately. Now, the messaging service, which is more robust and contains orders of magnitude more computational resources, will handle secure duplication and transmission of messages. Also removed is the need for public or private keys since the queue ID generated by the messaging service acts as a long, unique shared key. Guaranteed delivery of messages and maintaining privacy of data can easily be implemented, since exchanges will only deliver messages to

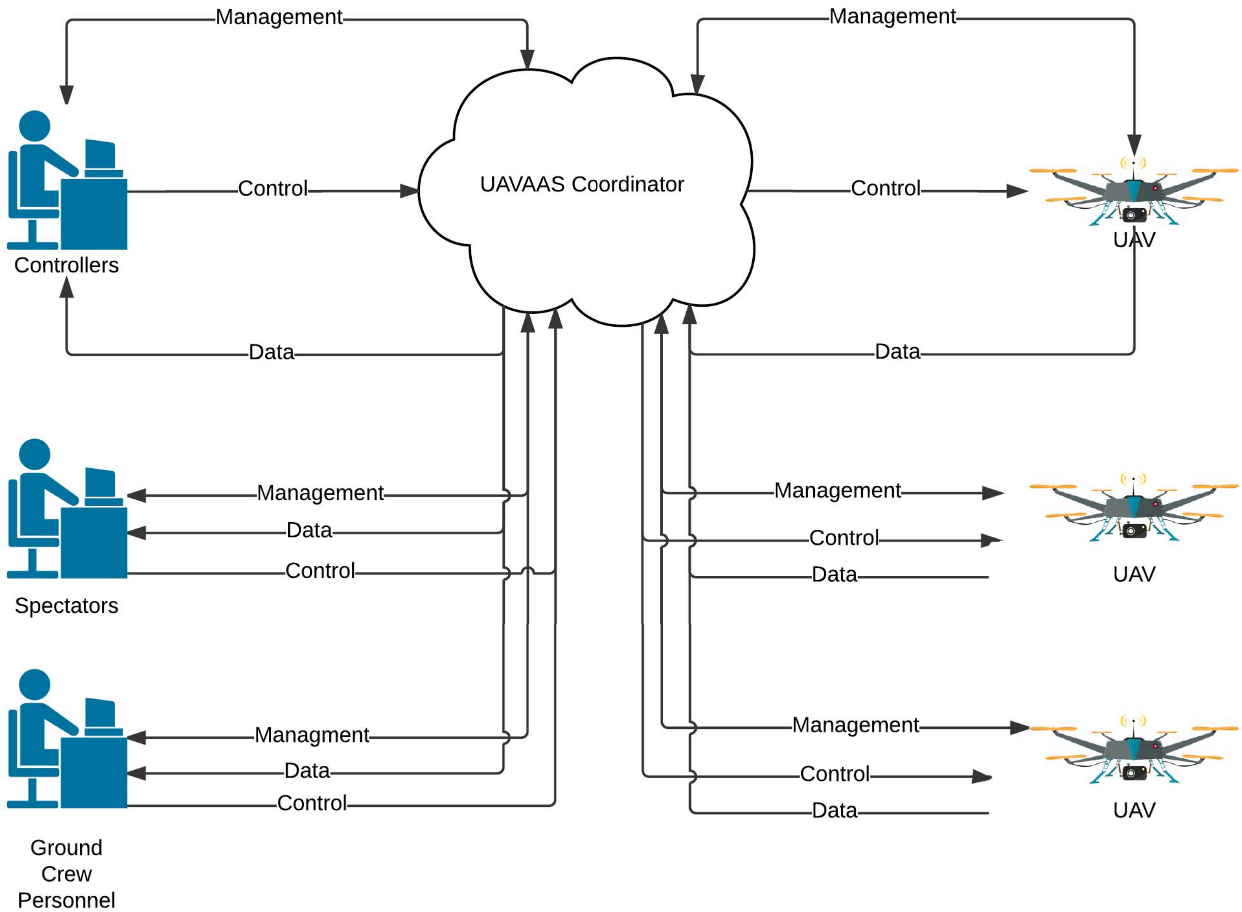

Fig. 3. High level system design

queues with the same topic as that labeled on the message.

B. UAV Design

The software components used to communicate with UAVaaS Coordinators are meant to be hardware agnostic. This allows a wide range of manufacturers to build compatible UAVs for UAVaaS providers without the need to build proprietary network interfaces. The key component of any UAV that connect to a service provider is an Adapter Module installed on a companion board. The UAV Adapter Module is responsible for: (1) contacting and updating the UAVaaS with status and availability information, (2) respond to new flight assignments, (3) receive and interpret control commands sent by operators, (4) package and send data messages to the UAVaaS Coordinator for distribution, (5) store public certificate and private key information needed for session negotiation, and (5) authenticate to a UAVaaS Cloud Provider.

UAVs will continue to send status updates to the UAVaaS Coordinator until a flight has been assigned. A flight assignment is sent when an operator has requested a scheduled mission to be started. Once a flight assignment is issued, the UAV will subscribe to relevant control exchanges on the messaging service using an assigned queue ID and topic label. The UAV will notify the management service that it is ready to begin the mission. From then on, updates will be sent, data such as video, photos, telemetry, warning and error messages will be published and control messages fetched from the control queues until a re-tasking or mission termination operation is performed.

Flight re-tasking is initiated by a higher priority entity such as emergency service personnel and generally follows the same process as tasking from the UAVs perspective. Once a UAV receives a re-task assignment it simply subscribes to the new control exchanges and records new topics to publish telemetry, warning/error and data messages. The UAV notifies the Management Service that reconfiguration is complete and continues flying as normal under new operator control.

VII. Justification For Approach

In this research we justify the use of a centralized cloud coordinator approach. This is accomplished by first identifying two alternative network topologies and cross-referencing them with their ability to satisfy requirements mentioned in Section V. Two primitive approaches analyzed are peer-to-peer networking and VPN networking.

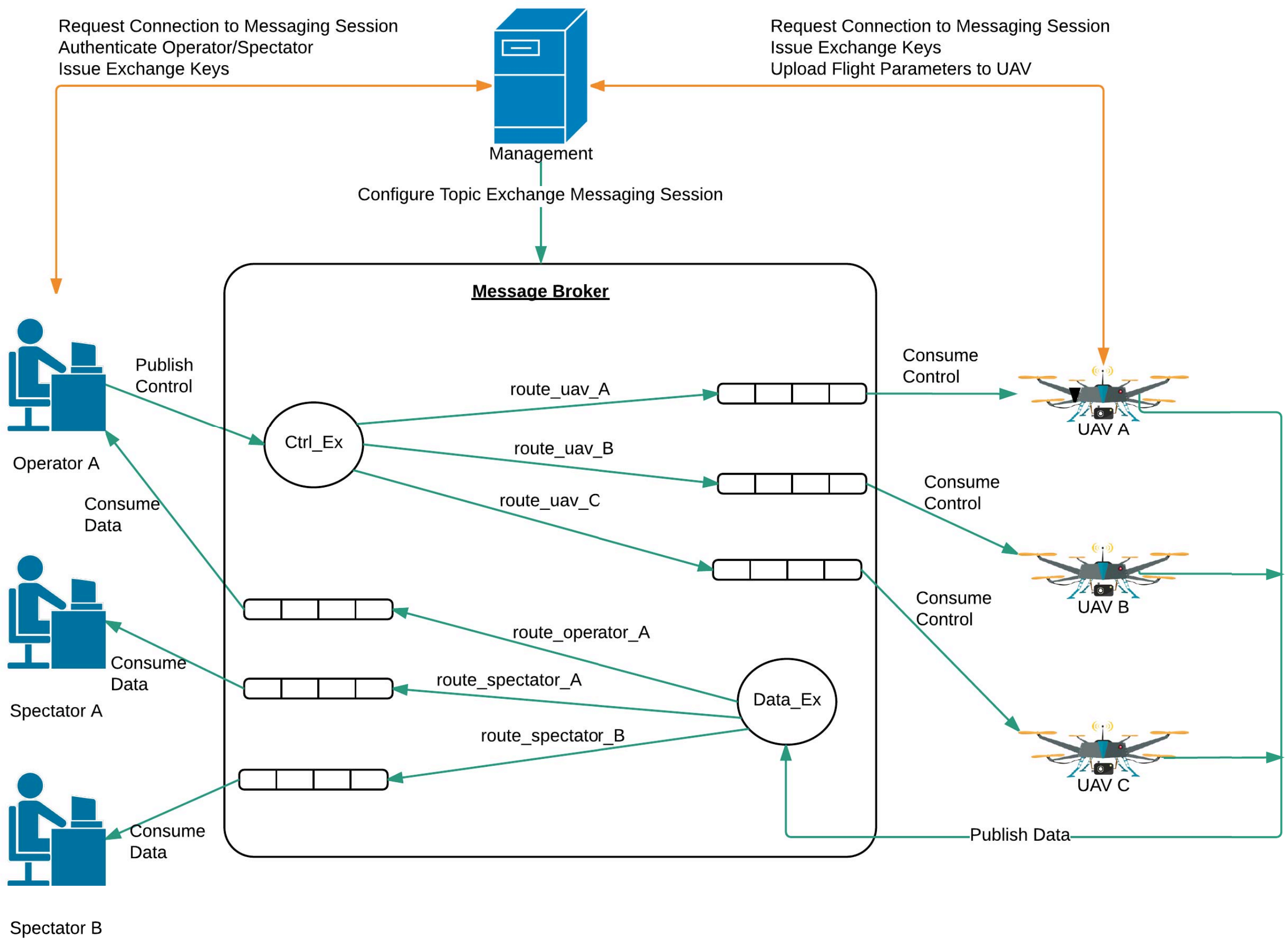

Fig. 4. High level system design

A. Peer-to-Peer Topology

Peer-to-Peer topology was chosen as a candidate because of its simplicity as illustrated in Figure 5, however underlying difficulties exist when trying to communicate between UAVs, spectators and controllers. Many issues are discovered which quickly made Point-to-Point communication unfeasible and insecure since it generated too much overhead for UAVs, spectators and controllers to handle on their own. Table I shows the results of this analysis and illustrates why point-to-point communication would not be a good solution.

B. VPN Topology

Virtual Private Networking is a technology that creates an encrypted connection over a less secure network, constructed using publicly accessible avenues over the internet. This allows multiple devices to connect and communicate in an isolated virtual network environment without worry of outside individuals eavesdropping on communication. Using this approach, dedicated servers can be attached to handle authentication and access control of users. Databases can store scheduling and

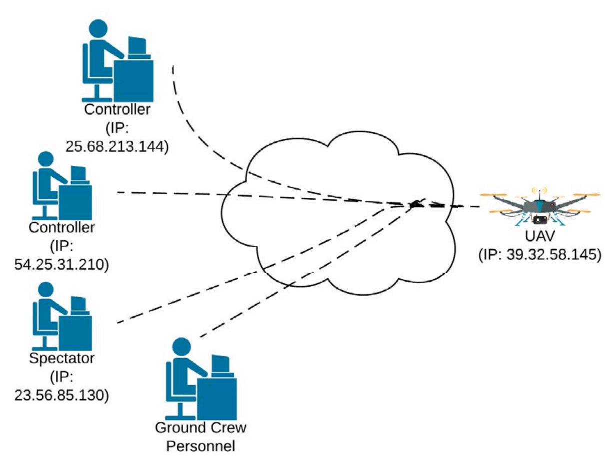

Fig. 5. Peer-to-Peer networking topolgoy with one UAV

reservation records that notifies UAVs of upcoming missions. Lastly, key management can create and store encryption keys

TABLE I
REQUIREMENTS FOR PEER-TO-PEER NETWORKING ALTERNATIVE

Requirement	Satisfied
Confidentiality and Privacy	NO
Integrity	NO
Authentication and Authorization	NO
Create and Manage Sessions	NO
Schedule and Reserve UAVs	NO
Retasking	NO
Scalability	NO

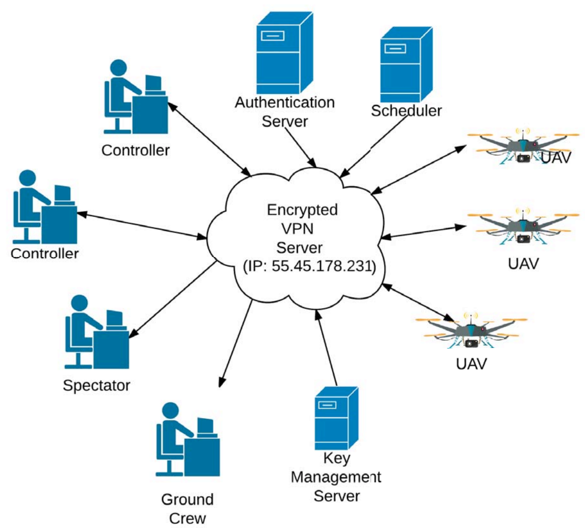

Fig. 6. VPN networking topology

TABLE II
REQUIREMENTS MET FOR VIRTUAL PRIVATE NETWORKING ALTERNATIVE

Requirement	Satisfied
Confidentiality and Privacy	YES
Integrity	NO
Authentication and Authorizatoin	YES
Create and Manage Sessions	NO
Schedule and Reserve UAVs	YES
Retasking	NO
Scalability	NO

to be used for sessions.

Although, virtual private networks provide secure communication to the outside world, it is still vulnerable to malicious attackers on the network. For example, two competing companies could launch denial of service attacks on each other, or intentionally intercept and corrupt messages.

VIII. RECOMMENDATIONS AND FUTURE WORK

This article presents a general description of what UAV as a Service is and how it could be designed and implemented. However before any future research and development of UAVaaS in the United States can begin, further research must be done to design UAVs that can intrinsically abide by rules and regulations set forth by the Federal Aviation Administration. Rules such as airspace restrictions, certifiable UAV designs, no-fly zone detection and monitoring and situational awareness algorithms should be considered before any implementation begins. Network performance and stress tests are considered outside the scope of this research. Since communication over cellular networks may be intermittent and unreliable at times, it is recommended that more research is placed on the amount of data consumption and bandwidth required for a UAVaaS system to be sustainable. If possible, Software Defined Networks should be researched and their applications assessed in a UAVaaS Cloud environment.

While UAVaaS demonstrates to reduce costs for customers and improve profit margins for UAV manufacturers, no cost-benefit analysis or detailed business models have been researched. It is recommended that more feasibility analysis is necessary on a business/marketing level, to determine optimal pricing for UAVaaS services and commissions awarded to UAVaaS vendors. Since UAVs that operate in a UAVaaS network typically travel outside line of sight of the operator, there is no way (apart from the UAVs camera) to detect and avoid nearby UAVs and aircraft. It is recommended that more research into this area be done before any further implementation of UAVaaS is carried out.

IX. CONCLUSION

This research has proposed and designed a new cloud orchestration framework that provides customers with affordable, on-demand access to multi-tenant UAVs that are supplied by a UAV as a Service Cloud Provider. Once implemented, UAVs (conforming to a UAVaaS specification) can be openly shared by businesses that wish to utilize commercial UAV services, without the legal and financial overheads associated with traditional upfront capital expenditure. On the other hand, third party vendors that faithfully supply the UAVaaS cloud provider with commercial grade UAVs, can make commission based profit, based on frequency and duration of UAV tasking. In order to secure the success of UAVaaS platform, a cloud provider must prioritize and satisfy the needs of third party vendors, operators, spectators, UAV hub ground crew, and emergency services personnel.

Design justification explains that both direct point-to-point communication and virtual private networking topologies between UAVs, operators and spectators are not sufficient to address major concerns on areas of security, safety and privacy. Instead a central, cloud service approach must be taken in which all communication is collaborated and routed through a coordinator.

A Representational State Transfer (REST) service for standard request-response queries coupled with an Advanced Message Queuing Protocol (AMQP) for messaging routing provides the simplest and most optimal design that takes into consideration factors of scalabiltiy and fault tolerance. In order for UAVs to connect to a UAVaaS Service Provider, vendors/manufacturers must take into consideration a standard design model of all UAV builds. 4G and wifi connectivity, Micro Air Vehicle Link (MAVLink) compatible autopilot controller and dedicated TCP/IP to MAVLink adapter must be incorporated into every build just to name a few.

With the year 2016 being hailed as ”the dawn of the drone age” [7], and commercial development of UAVs now in full swing, focus needs to be placed on designing efficient, cloud-friendly ecosystems that will allow UAVs to collaboratively work together for the greater good of humankind. UAV as a Service is one such ecosystem.

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